

Project Euler 15 - Lattice Paths

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Assume that we are only able to move right and down. In a 2×2 grid, to get from the top left to the bottom right we need to make 2 right moves and 2 down moves. Once we choose the down moves (resp. right moves) the right moves (resp. down moves) are automatically chosen. So from the 4 total moves we only have to choose 2. There are

$$\binom{4}{2} = 6 \text{ ways to do it}$$

Thus, 6 possible routes.

For a $n \times n$ grid, we need a total of $2n$ moves, n right and n down. Again, the number of possible paths equals the number of possible choices of the down (or the right) moves. Thus, there are

$$\binom{2n}{n} \text{ possible paths.}$$

For the 20×20 grid case, the number of possible routes is

$$\boxed{\binom{40}{20} = 137846528820}$$